

# WASP-12b

R-0586

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-12\_240122 · created 2026-07-24T20:46:16+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460331.7003 +/- 0.0011 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.152 +/- 0.045                 |
| Transit depth (Rp/Rs)^2                           | 2.32 +/- 1.36 [%]               |
| Orbital Inclination (inc)                         | 86.39 +/- 2.92                  |
| Airmass coefficient 1 (a1)                        | 13.12 +/- 7.93                  |
| Airmass coefficient 2 (a2)                        | -2.71 +/- 1.49                  |
| Scatter in the residuals of the lightcurve fit is | 69.23 %                         |
| Best Comparison Star                              | #2 - [418, 141]                 |
| Optimal Aperture                                  | 1.7                             |
| Optimal Annulus                                   | 5.0                             |
| Transit Duration (day)                            | 0.1163 +/- 0.0035               |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.152 ± 0.045 vs 0.1178 (deviation 0.76σ) |
| Duration fit vs published | 0.1163 d vs 0.125 d (deviation 2.49σ)     |
| Mid-time O–C              | 1.1 min                                   |
| Depth SNR                 | 3.4                                       |
| Residual scatter          | 69.23 %                                   |

## Light curve

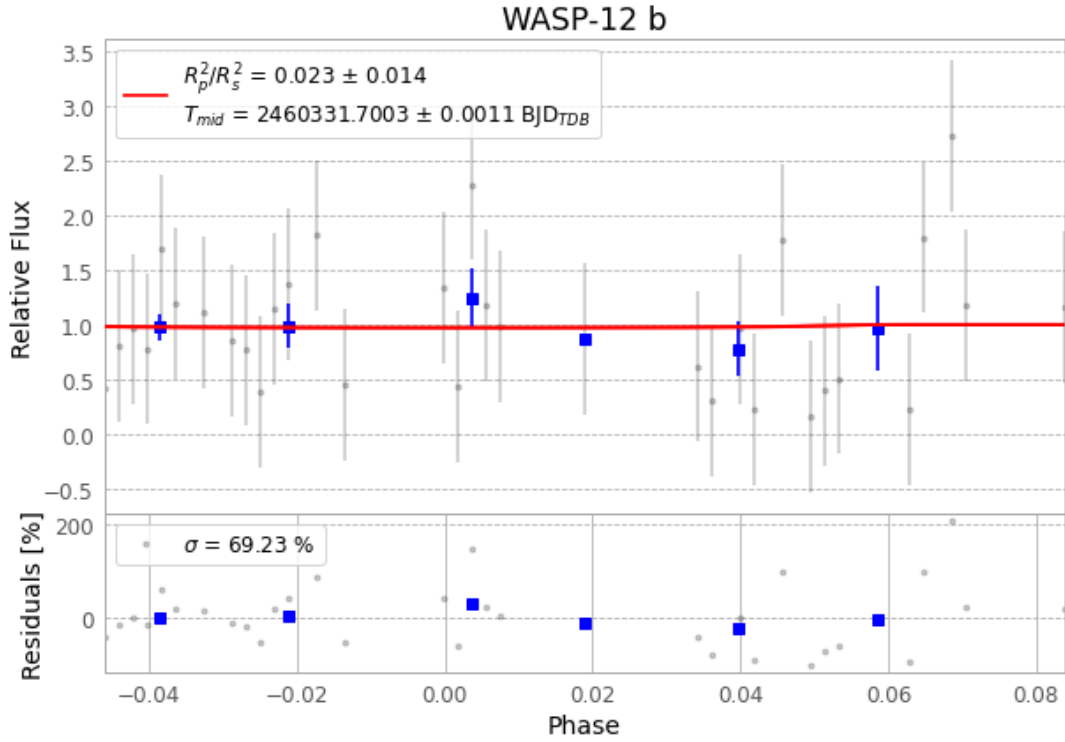


Figure 1 — FinalLightCurve\_WASP-12 b\_2024-01-21.png (SHA-256 6ff049eba0569ae7...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [309, 359], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 4c8fb670cbc1398d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit b365494

## Data provenance

78 FITS frames (51 MB), WASP-12240122032415.FITS → WASP-12240122071515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-240122.log (5864714ab9b6...), FinalLightCurve\_WASP-12 b\_2024-01-21.png (6ff049eba056...), FinalLightCurve\_WASP-12 b\_2024-01-21.csv (0ea3844b3b0f...)

## Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 20:46Z.